⚙️ Settings

LIME samples

400

SHAP background size

30

GradCAM overlay α

0.45

- [x] Run LIME
- [x] Run SHAP (slow ~30s)

Classes

· Chickenpox

· Cowpox

· HFMD

· Healthy

· Measles

· Monkeypox

🔬 MorphNN — Viral Exanthem Classifier

Joint Morphology + CNN model (C21) · Macro F1 = **0.9977** · Explains predictions with GradCAM, LIME, and SHAP.

Upload a skin lesion image

Upload 200MB per file • JPG, JPEG, PNG, BMP

👆 Upload a skin lesion image to classify it and see explainability maps.

Supported conditions: Chickenpox · Cowpox · HFMD · Healthy · Measles · Monkeypox

How it works

Step	Method	Purpose
1️⃣	Morphological feature extraction	16 lesion descriptors from OpenCV
2️⃣	Joint CNN + Morph inference	MobileNetV3 + morph branch, end-to-end
3️⃣	GradCAM	Which image regions drove the decision
4️⃣	LIME	Which superpixels support the prediction
5️⃣	SHAP	Which morphological features mattered most

Model performance: Macro F1 = **0.9977** (1,134 test images) — beats standalone CNN (0.9946)